M339G/M389G Syllabus

M339G/M389G: Predictive Analytics - Spring 2026 - Syllabus

COURSE-SPECIFIC INFORMATION

Welcome to M339G/M389G! Here is some information and some ground rules. Read carefully and let me know if there is anything unclear **by the twelfth day of classes, i.e., January 28th, 2025.**

Basic information

Course number. M339G/M389G (unique: 56750/57095)

Course meets. MWF 10am - 10:50am in PMA 5.122

Instructor. Milica Čudina; my office is PMA 13.142 (2515 Speedway, Austin, TX 78712).

Email. It's best to use Canvas to email me; my email address is mcudina@math.utexas.edu.

Office Hours. MWF 11am-11:50am in PMA 13.142.

Course info

Course description. This course provides an introduction to predictive modelling starting. The emphasis will be on fitting suitable models in supervised learning both in regression and classification settings. The course includes resampling methods, simple linear regression, multiple linear regression, cross-validation, splines and tree-based methods as well as several specific classification methods. Some unsupervised learning methods are discussed: principal components analysis and **K** − means clustering.

Learning outcomes.

- Contrasting supervised and unsupervised learning.
- Contrasting regression and classification problems.
- Assessing the strength of resampling procedures and their relationship to classical techniques.
- Transferring ideas of linear algebra to random variable manipulations.
- Differentiating notions of accuracy.
- Assessing model accuracy in specific settings.
- Generalizing the fundamental models to more sophisticated ones.
- Recognizing ethical connotations of multiple hypothesis testing (e.g., in the context of multiple linear regression).
- Building statistical self-confidence.

Prerequisites. The formal prerequisite is the grade C- or better in M378K **and** M341 (or M340L, M340L-CS).

Students are also assumed to have prior programming experience, preferably with R.

Lectures online. This class is using the *Lectures Online* recording system. This system records the audio and video material presented in class for you to review after class. Links for the recordings will appear in the *Lectures Online* tab on the Canvas page for this class. You will find this tab along the left side navigation in Canvas.

To review a recording, simply click on the Lectures Online navigation tab and follow the instructions presented to you on the page. You can learn more about how to use the Lectures Online system at http://sites.la.utexas.edu/lecturesonline/students/how-to-access-recordings/.

You can find additional information about Lectures Online at: https://sites.la.utexas.edu/lecturesonline/.

Class format and attendance. Attendance for the purposes of grading will **not** be taken. However, regular attendance is strongly recommended. In case you need to be absent, you are responsible for covering the missed material independently. Class notes will be provided on the course website. There will be **no** synchronous online option for this course. You are strongly encouraged to stay home if you are sick or contagious, not only to stop the spread of disease but also to promote your personal wellness. I view this class as a community of learners. We cannot learn effectively when we are ill. Please, take care of yourselves and your classmates.

Here are some university resources on COVID-19.

If **students** are isolating, too sick to attend class, or experiencing another type of absence, they should:

- contact the Student Emergency Services immediately, and
- email the instructor as soon as they feel well enough to do so.

If the **instructor** is isolating, or too sick to attend class, she will do her best to change class modality to Zoom (or find an alternative instructor if the situation calls for such drastic measures and if it's possible).

The class meetings consist of interactive lectures, coding demonstrations, and problem solving. In short, the course will incorporate a lot of active learning in class. Thus, if you miss class, you miss out on these learning opportunities. Please, come to class as much as possible.

Textbook. "An Introduction to Statistical Learning (with applications in R)" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani (Second edition)

Required devices. You will need access to a computer to be able to work on projects and homework.

Online resources.

1. **Course website**: https://mcudina.github.io/page/M339G/M339G.html. I recommend bookmarking this course site in your default browser for easy access.
2. **Canvas** will be used in this course to keep track of grades and for communication purposes. The students are responsible for the content of these announcements. The easiest way not to miss any is to turn on (i.e., not turn off) *Announcements* in their account's *Notification* menu.
3. **Ed Discussion** will be used for informal class discussion. The system is highly catered to getting you help fast and efficiently from classmates and myself. Rather than emailing questions to the instructor, I encourage you to post your questions on Ed Discussion. It is accessible via the menu on the left-hand side in Canvas.

Sharing of Course Materials is Prohibited. No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. Any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

Class Recordings. Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Assessment and grading

A note on departmental policy. To address concerns on academic integrity and student development, the Department of Mathematics is requesting that midterms and final exams be in-person and proctored.

Homework assignments. Homework assignments will be available on the course website or in Canvas. You will be uploading your solutions using Canvas. Please, have your solutions in order and number the pages. Having read and understood this First-Day Handout in its entirety will count as the zeroth homework assignment. To get the credit, read this entire document with understanding by the homework deadline. Not handing in this assignment does not exempt you from abiding by this First-Day Handout. **The lowest three homework scores will be dropped. The homework assignments and their due dates will be announced as the term progresses.**

Projects. There will be **three** in-person group projects. The formulations and due dates for the group projects will be available on the course website. The nature and content of the projects will be described in more detail as new techniques are introduced. However, every group-project will be done as part of a self-assigned group of students and require critical thinking and drawing logical conclusions.

The projects are designed to include open-ended problems which do not necessarily have a unique final answer. For that reason, there is **no** checklist-type rubric for the projects. In grading, emphasis will be placed on professional presentation, appropriate choice of method, and accurate interpretation of results. If you submit simply the code and its output, you will get at most 30% of the total available points.

No late projects or homework are accepted except in dire circumstances at the sole discretion of the instructor.

In-term exams. There will be two in-term exams. Both will be individual and conducted in-person in our classroom.

The exam coverage will be shared on the course website ahead of the exam itself. Likewise, you will be given access to practice questions. The exams drew heavily on actuarial exams SRM, MAS-I and MAS-II while maintaining intellectual robustness you are familiar with from prior math classes. As such, they consist of a mixture of definitions, qualitative questions, free-response problems, and multiple-choice questions.

The exams will all be handwritten on paper provided by the instructor. You will be allowed to use calculators and writing implements. In particular, your laptops will **not** be allowed in exams.

If you miss an exam due to illness or other extenuating circumstances, the final exam will take the weight of the in-term exam you missed (on top of its original weight). If you miss more than one in-term exam, you are strongly encouraged to seek assistance from the Office of the Dean of Students to explore what your options are in such a dire situation.

The Final Exam. The Pre-Final Grade. The pre-final grade is composed as follows:

Assignment	Percentage of final grade
Homework	15%
Group projects	45% (15% each)
In-term exams	40% (20% each)

If you are satisfied with your course grade (see table below) based on your pre-final performance, you can opt out of the final exam by selecting TRUE in the *Final-exam opt out* Canvas quiz. If you missed any of the in-term exams, you are required to take the final exam. If you do not opt out of the final exam, your final-exam score **will** be incorporated in the calculation of the final score in the course as described below.

The Final Exam. The individual final exam is going to be comprehensive. That means that any material covered in class or assigned as reading can (and probably will) appear. Our comprehensive final exam will take place in our regular classroom on **Friday, May 1st, 1:00 pm-3:00 pm**.

Final grade. The final grade is composed as follows:

Assignment	Percentage of final grade
Homework	14%
Group projects	36% (12% each)
In-term exams	30% (15% each)
The final exam	20%

There is *no curve* in this class and the letter grades are assigned according to the following table:

A	A-	B+	B	B-	C+	C	C-	D+	D	D-
94-100	90-94	86 - 90	82 - 86	78 - 82	74 - 78	70 - 74	65 - 70	60 - 65	55 - 60	50 - 55

GENERAL, UNIVERSITY- or STATE-MANDATED INFORMATION

Drop dates. The procedure/consequences are different, depending on whether you drop before or after the 6th day of classes *(01/20)*, and then, before or after the *main drop (Q-drop) date (04/15)*. (See https://registrar.utexas.edu/calendars/25-26 for details)

Students with Disabilities. The University of Texas at Austin provides upon request appropriate academic accommodations for qualified students with disabilities. If you have a documented disability and you need specific support as a result of your disability, please let me know as soon as possible, but definitely within the first 3 weeks of class. For more information, contact the Office of the Dean of Students at 471-6259, 471-4641 (TTY), 1-866-329- 3986 (video phone) or go to https://disability.utexas.edu/

Counseling and mental health. Counseling and other mental-health services are available from **Counseling and Mental Health Center**, Student Services Bldg (SSB), 5th Floor (hours: M-F 8am–5pm, phone: 512 471 3515, web: http://www.cmhc.utexas.edu)

Religious holy days. Religious holy days sometimes conflict with class and examination schedules. Sections 51.911 and 51.925 of the Texas Education Code relate to absences by students and instructors for observance of religious holy days.

Section 51.911 states that a student who misses an examination, work assignment, or other project due to the observance of a religious holy day must be given an opportunity to complete the work missed within a reasonable time after the absence, provided that they have properly notified each instructor.

It is the policy of The University of Texas at Austin that the student must notify each instructor at least fourteen days prior to the classes scheduled on dates he or she will be absent to observe a religious holy day. For religious holidays that fall within the first two weeks of the semester, the notice should be given on the first day of the semester. The student may not be penalized for these excused absences but the instructor may appropriately respond if the student fails to complete satisfactorily the missed assignment or examination within a reasonable time after the excused absence.

Title IX Reporting/SB 212. Texas *Senate Bill 212* requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Your instructor in a **mandatory reporter**. By law, your instructor must be fired if she does not report. Our Student Ombuds is confidential. Additionally, if you wish to speak with someone who can provide support **without** making an official report to the university, contact a confidential case manager by emailing advocate@austin.utexas.edu. Case managers can also provide support, resources, and accommodations for pregnant, nursing, and parenting students.

Sanger Learning Center. All students are welcome to take advantage of Sanger Center's classes and workshops, private learning specialist appointments, peer academic coaching, and tutoring for more than 70 courses in 15 different subject areas. For more information, please visit https://undergraduates.utexas.edu/tutoring-academic-assistance/sanger-learning-center or call 512-471-3614 (JES A332).

Important Safety Information. Here is a comprehensive list of Safety, Health and Security Resources

Occupants of buildings on The University of Texas at Austin campus are required to evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.

- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
- In the event of an evacuation, follow the instruction of faculty or class instructors. Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.
- Link to information regarding emergency evacuation routes and emergency procedures can be found at: http://www.utexas.edu/emergency

Academic (dis)Honesty. Students who violate University rules on academic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. Since such dishonesty harms the individual, all students, and the integrity of the University, policies on academic dishonesty will be strictly enforced. For further information, please visit the Student Conduct and Academic Integrity website at: http://deanofstudents.utexas.edu/conduct For a more detailed document, please consult: https://catalog.utexas.edu/general-information/appendices/appendix-c/student-conduct-and-academic-integrity/ Please, pay particular attention to the section on *plagiarism*.

This syllabus is subject to change. If you have to miss class, please make sure to check in with a classmate to learn of any updates that were made in your absence.

The SCHEDULE of CLASSES

Date	Weekday	Topic
01/12/2026	Mon	Orientation. Rmd.
01/14/2026	Wed	Estimation. Bias. MSE.
01/16/2026	Fri	Resampling Methods (Sampling Distribution and Univariate Bootstrap).
01/21/2026	Wed	Resampling Methods (Bootstrap, cont'd).
01/23/2026	Fri	Prediction and Inference.
01/26/2026	Mon	More on prediction and inference.
01/28/2025	Wed	Simple Linear Regression.
01/30/2026	Fri	Resampling Methods (Sampling Distribution of the Slope Coefficient).
02/02/2026	Mon	Simple Regression: Cross Validation.
02/04/2026	Wed	Categorical Predictors. Multiple Linear Regression.
02/06/2026	Fri	Multiple Linear Regression (cont'd).
02/09/2026	Mon	Multiple Linear Regression (cont'd). Splines.
02/11/2026	Wed	Splines (cont'd).
02/13/2026	Fri	The Trade-Off Between Prediction Accuracy and Model Interpretability.
02/16/2026	Mon	Introduction to Classification.
02/18/2026	Wed	Regression vs Classification. Logistic Regression.
02/20/2026	Fri	More on Logistic Regression.
02/23/2026	Mon	Logistic regression with multiple response categories.
02/25/2026	Wed	K-Nearest Neighbors. Collinearity.
02/27/2026	Fri	Supervised vs Unsupervised Learning. K-Means Clustering.
03/02/2026	Mon	Singular Value Decomposition.
03/04/2026	Wed	*In-Term One*
03/06/2026	Fri	Principal Component Analysis (PCA).
03/09/2026	Mon	Principal Component Regression. PCA and Clustering.
03/11/2026	Wed	Linear Discriminant Analysis.
03/13/2026	Fri	Bivariate Normal Distribution.
03/23/2026	Mon	Bivariate Normal Distribution (cont'd).
03/25/2026	Wed	Multivariate Normal Distribution.
03/27/2026	Fri	Quadratic Discriminant Analysis.
03/30/2026	Mon	LDA, QDA, naive Bayes.
04/01/2026	Wed	Tree-Based Regression.
04/03/2026	Fri	Pruning.
04/06/2026	Mon	Tree-Based Classification.
04/08/2026	Wed	Bagging.
04/10/2026	Fri	Random Forest. Boosting.
04/13/2026	Mon	Lines, Planes, Hyperplanes.
04/15/2026	Wed	*In-Term Two*
04/17/2026	Fri	Maximal Margin Classifier.
04/20/2026	Mon	Support Vector Classifiers. Support Vector Machines.
04/22/2026	Wed	A Quick Review of Hypothesis Testing.
04/24/2026	Fri	The Challenge of Multiple Testing.
04/27/2026	Mon	The False Discovery Rate.